



Advanced Linear Algebra: Foundations to Frontiers

Robert van de Geijn, Margaret Myers

2.3.1 The Singular Value Decomposition Theorem



02.3.1 The SVD

fit width



The following is probably the most important result in linear algebra:

Theorem 2.3.1.1. Singular Value Decomposition Theorem. *Given $A \in \mathbb{C}^{m \times n}$ there exist unitary $U \in \mathbb{C}^{m \times m}$, unitary $V \in \mathbb{C}^{n \times n}$, and $\Sigma \in \mathbb{R}^{m \times n}$ such that $A = U\Sigma V^H$. Here*

$$\Sigma = \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right) \text{ with } \Sigma_{TL} = \begin{pmatrix} \sigma_0 & 0 & \cdots & 0 \\ 0 & \sigma_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{r-1} \end{pmatrix} \quad (2.3.1)$$

and $\sigma_0 \geq \sigma_1 \geq \cdots \geq \sigma_{r-1} > 0$. The values $\sigma_0, \dots, \sigma_{r-1}$ are called the singular values of matrix A . The columns of U and V are called the left and right singular vectors, respectively.

Recall that in our notation a 0 indicates a matrix of vector "of appropriate size" and that in this setting the zero matrices in (2.3.1) may be 0×0 , $(m - r) \times 0$, and/or $0 \times (n - r)$.

Before proving this theorem, we are going to put some intermediate results in place.

Remark 2.3.1.2. As the course progresses, we will notice that there is a conflict between the notation that explicitly exposes indices, e.g.,

$$U = (u_0 \quad u_1 \quad \cdots \quad u_{n-1})$$

and the notation we use to hide such explicit indexing, which we call the FLAME notation, e.g.,

$$U = \left(U_0 \mid u_1 \quad U_2 \right).$$

The two linked by

$$\left(\underbrace{u_0 \quad u_{k-1}}_{U_0} \mid \underbrace{u_k}_{u_1} \quad \underbrace{u_{k+1} \quad u_{n-1}}_{U_2} \right).$$

In algorithms that use explicit indexing, k often is the loop index that identifies where in the matrix or vector the algorithm currently has reached. In the FLAME notation, the index 1 identifies that place. This creates a conflict for the two distinct items that are both indexed with 1, e.g., u_1 in our example here. It is our experience that learners quickly adapt to this and hence have not tried to introduce even more notation that avoids this conflict. In other words: you will almost always be able to tell from context what is meant. The following lemma and its proof illustrate this further.

Lemma 2.3.1.3. Given $A \in \mathbb{C}^{m \times n}$, with $1 \leq n \leq m$ and $A \neq 0$ (the zero matrix), there exist unitary matrices $\tilde{U} \in \mathbb{C}^{m \times m}$ and $\tilde{V} \in \mathbb{C}^{n \times n}$ such that

$$A = \tilde{U} \left(\frac{\sigma_1}{0} \mid \frac{0}{B} \right) \tilde{V}^H, \text{ where } \sigma_1 = \|A\|_2.$$

Proof.

Hopefully you can see where this is going: If one can recursively find that $B = U_B \Sigma_B V_B^H$, then

$$\begin{aligned}
A &= \tilde{U} \left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & B \end{array} \right) \tilde{V}^H \\
&= \tilde{U} \left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & U_B \Sigma_B V_B^H \end{array} \right) \tilde{V}^H \\
&= \tilde{U} \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & U_B \end{array} \right) \left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & \Sigma_B \end{array} \right) \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & V_B^H \end{array} \right) \tilde{V}^H \\
&= \underbrace{\tilde{U} \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & U_B \end{array} \right)}_U \underbrace{\left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & \Sigma_B \end{array} \right)}_\Sigma \underbrace{\left(\tilde{V} \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & V_B^H \end{array} \right) \right)^H}_{V^H}.
\end{aligned}$$

The next exercise provides the insight that the values on the diagonal of Σ will be ordered from largest to smallest.

Homework 2.3.1.1. Let $A \in \mathbb{C}^{m \times n}$ with $A = \left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & B \end{array} \right)$ and assume that

$$\|A\|_2 = \sigma_1.$$

ALWAYS/SOMETIMES/NEVER: $\|B\|_2 \leq \sigma_1$.

▼ Solution

We will employ a proof by contradiction. Assume that $\|B\|_2 > \sigma_1$. Then there exists a vector z with $\|z\|_2 = 1$ such that $\|B\|_2 = \|Bz\|_2 = \max_{\|x\|_2=1} \|Bx\|_2$. But then

$$\begin{aligned}
& \|A\|_2 \\
&= \text{< definition >} \\
& \max_{\|x\|_2=1} \|Ax\|_2 \\
&\geq \text{< pick a specific vector with 2-norm equal to one >} \\
& \left\| A \begin{pmatrix} 0 \\ z \end{pmatrix} \right\|_2 \\
&= \text{< instantiate } A \text{ >} \\
& \left\| \left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & B \end{array} \right) \begin{pmatrix} 0 \\ z \end{pmatrix} \right\|_2 \\
&= \text{< partitioned matrix-vector multiplication >} \\
& \left\| \begin{pmatrix} 0 \\ Bz \end{pmatrix} \right\|_2 \\
&= \text{< } \left\| \begin{pmatrix} y_0 \\ y_1 \end{pmatrix} \right\|_2^2 = \|y_0\|_2^2 + \|y_1\|_2^2 \text{ >} \\
& \|Bz\|_2 \\
&= \text{< assumption about } z \text{ >} \\
& \|B\|_2 \\
&> \text{< assumption >} \\
& \sigma_1.
\end{aligned}$$

which is a contradiction.

Hence $\|B\|_2 \leq \sigma_1$.

✎ We are now ready to prove the Singular Value Decomposition Theorem.

✎ Proof of Singular Value Decomposition Theorem for $n \leq m$.

We will prove this for $m \geq n$, leaving the case where $m \leq n$ as an exercise.

Proof by induction: Since $m \geq n$, we select m to be arbitrary and induct on n .

- Base case: $n = 1$.

In this case $A = (a_1)$ where $a_1 \in \mathbb{C}^m$ is its only column.

Case 1: $a_1 = 0$ (the zero vector).

Then

$$A = (0) = \underbrace{I_{m \times m}}_U \left(\begin{array}{c|c} & \\ \hline & 0 \end{array} \right) \underbrace{I_{1 \times 1}}_{V^H}$$

so that $U = I_{m \times m}$, $V = I_{1 \times 1}$, and Σ_{TL} is an empty matrix.

Case 2: $a_1 \neq 0$.

Then

$$A = (a_1) = (u_1)(\|a_1\|_2)$$

where $u_1 = a_1/\|a_1\|_2$. Choose $U_2 \in \mathbb{C}^{m \times (m-1)}$ so that $U = \left(u_1 \mid U_2 \right)$ is unitary. Then

$$\begin{aligned} A &= (a_1) \\ &= (u_1)(\|a_1\|_2) \\ &= \left(u_1 \mid U_2 \right) \left(\begin{array}{c|c} \|a_1\|_2 & \\ \hline 0 & \end{array} \right) (1)^H \\ &= U \Sigma V^H, \end{aligned}$$

where

- $U = \left(u_1 \mid U_2 \right)$,
- $\Sigma = \left(\begin{array}{c|c} \Sigma_{TL} & \\ \hline 0 & \end{array} \right)$ with $\Sigma_{TL} = (\sigma_1)$ and $\sigma_1 = \|a_1\|_2 = \|A\|_2$
- $V = (1)$.

- Inductive step:

Assume the result is true for matrices with $1 \leq k$ columns. Show that it is true for matrices with $k + 1$ columns.

Let $A \in \mathbb{C}^{m \times (k+1)}$ with $1 \leq k < n$.

Case 1: $A = 0$ (the zero matrix)

Then

$$A = I_{m \times m} \left(\begin{array}{c|c} & \\ \hline 0_{m \times (k+1)} & \end{array} \right) I_{(k+1) \times (k+1)}$$

so that $U = I_{m \times m}$, $V = I_{(k+1) \times (k+1)}$, and Σ_{TL} is an empty matrix.

Case 2: $A \neq 0$.

Then $\|A\|_2 \neq 0$. By [Lemma 2.3.1.3](#), we know that there exist unitary

$\tilde{U} \in \mathbb{C}^{m \times m}$ and $\tilde{V} \in \mathbb{C}^{(k+1) \times (k+1)}$ such that $A = \tilde{U} \left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & B \end{array} \right) \tilde{V}^H$ with

$\sigma_1 = \|A\|_2$.

By the inductive hypothesis, there exist unitary $\check{U}_B \in \mathbb{C}^{(m-1) \times (m-1)}$,

unitary $\check{V}_B \in \mathbb{C}^{k \times k}$, and $\check{\Sigma}_B \in \mathbb{R}^{(m-1) \times k}$ such that $B = \check{U}_B \check{\Sigma}_B \check{V}_B^H$ where

$$\tilde{\Sigma}_B = \left(\begin{array}{c|c} \tilde{\Sigma}_{TL} & 0 \\ \hline 0 & 0 \end{array} \right), \tilde{\Sigma}_{TL} = \text{diag}(\sigma_2, \dots, \sigma_{r-1}), \text{ and}$$

$$\sigma_2 \geq \dots \geq \sigma_{r-1} > 0.$$

Now, let

$$U = \tilde{U} \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & \tilde{U}_B \end{array} \right), V = \tilde{V} \left(\begin{array}{c|c} 1 & 0 \\ \hline 0 & \tilde{V}_B \end{array} \right), \text{ and } \Sigma = \left(\begin{array}{c|c} \sigma_1 & 0 \\ \hline 0 & \tilde{\Sigma}_B \end{array} \right).$$

(There are some really tough to see "checks" in the definition of U , V , and Σ !!) Then $A = U\Sigma V^H$ where U , V , and Σ have the desired properties. Key here is that $\sigma_1 = \|A\|_2 \geq \|B\|_2$ which means that $\sigma_1 \geq \sigma_2$.

- By the Principle of Mathematical Induction the result holds for all matrices $A \in \mathbb{C}^{m \times n}$ with $m \geq n$.

Homework 2.3.1.2. Let $\Sigma = \text{diag}(\sigma_0, \dots, \sigma_{n-1})$. ALWAYS/SOMETIMES/NEVER:

$$\|\Sigma\|_2 = \max_{i=0}^{n-1} |\sigma_i|.$$

▼ [Answer](#) ▼ [Solution](#)

Yes, you have seen this before, in [Homework 1.3.5.1](#). We repeat it here because of its importance to this topic.

$$\begin{aligned} \|\Sigma\|_2^2 &= \max_{\|x\|_2=1} \|\Sigma x\|_2^2 \\ &= \max_{\|x\|_2=1} \left\| \begin{pmatrix} \sigma_0 & 0 & \cdots & 0 \\ 0 & \sigma_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{n-1} \end{pmatrix} \begin{pmatrix} \chi_0 \\ \chi_1 \\ \vdots \\ \chi_{n-1} \end{pmatrix} \right\|_2^2 \\ &= \max_{\|x\|_2=1} \left\| \begin{pmatrix} \sigma_0 \chi_0 \\ \sigma_1 \chi_1 \\ \vdots \\ \sigma_{n-1} \chi_{n-1} \end{pmatrix} \right\|_2^2 \\ &= \max_{\|x\|_2=1} \left[\sum_{j=0}^{n-1} |\sigma_j \chi_j|^2 \right] \\ &= \max_{\|x\|_2=1} \left[\sum_{j=0}^{n-1} [|\sigma_j|^2 |\chi_j|^2] \right] \\ &\leq \max_{\|x\|_2=1} \left[\sum_{j=0}^{n-1} [\max_{i=0}^{n-1} |\sigma_i|^2 |\chi_j|^2] \right] \\ &= \max_{\|x\|_2=1} \left[\max_{i=0}^{n-1} |\sigma_i|^2 \sum_{j=0}^{n-1} |\chi_j|^2 \right] \\ &= (\max_{i=0}^{n-1} |\sigma_i|)^2 \max_{\|x\|_2=1} \|x\|_2^2 \\ &= (\max_{i=0}^{n-1} |\sigma_i|)^2. \end{aligned}$$

so that $\|\Sigma\|_2 \leq \max_{i=0}^{n-1} |\sigma_i|$.

Also, choose j so that $|\sigma_j| = \max_{i=0}^{n-1} |\sigma_i|$. Then

$$\|\Sigma\|_2 = \max_{\|x\|_2=1} \|\Sigma x\|_2 \geq \|\Sigma e_j\|_2 = \|\sigma_j e_j\|_2 = |\sigma_j| \|e_j\|_2 = |\sigma_j| = \max_{i=0}^{n-1} |\sigma_i|.$$

so that $\max_{i=0}^{n-1} |\sigma_i| \leq \|\Sigma\|_2 \leq \max_{i=0}^{n-1} |\sigma_i|$, which implies that $\|\Sigma\|_2 = \max_{i=0}^{n-1} |\sigma_i|$.

ALWAYS

Now prove it.

Homework 2.3.1.3. Assume that $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ are unitary matrices. Let $A, B \in \mathbb{C}^{m \times n}$ with $B = U A V^H$. Show that the singular values of A equal the singular values of B .

▼ Solution

Let $A = U_A \Sigma_A V_A^H$ be the SVD of A . Then $B = U U_A \Sigma_A V_A^H V^H = (U U_A) \Sigma_A (V V_A)^H$ where both $U U_A$ and $V V_A$ are unitary. This gives us the SVD for B and it shows that the singular values of B equal the singular values of A .

Homework 2.3.1.4. Let $A \in \mathbb{C}^{m \times n}$ with $n \leq m$ and $A = U \Sigma V^H$ be its SVD.

ALWAYS/SOMETIMES/NEVER: $A^H = V \Sigma^T U^H$.

▼ Answer ▼ Solution

$$A^H = (U \Sigma V^H)^H = (V^H)^H \Sigma^T U^H = V \Sigma^T U^H$$

since Σ is real valued. Notice that Σ is only "sort of diagonal" (it is possibly rectangular) which is why $\Sigma^T \neq \Sigma$.

ALWAYS

Homework 2.3.1.5. Prove the Singular Value Decomposition Theorem for $m \leq n$.

▼ Hint ▼ Solution

Let $B = A^H$. Since it is $n \times m$ with $n \geq m$ its SVD exists: $B = U_B \Sigma_B V_B^H$. Then $A = B^H = V_B \Sigma_B^T U_B^H$ and hence $A = U \Sigma V^H$ with $U = V_B$, $\Sigma = \Sigma_B^T$.

and $V = U_B$.

Consider the SVD of $B = A^H$

 ***I believe the following video has material that is better presented in second video of 2.3.2.***

02.3.1 SVD and change of basis

fit width



